

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2- Industry Based Problem Solving

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4	Assessment:	Assessment Tool 2- Industry Based Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	B C Yugesh	Reg. No.:	192424282
Section / Batch:	2024	Date:	24/08/2026

Code of Conduct

I **B C Yugesh (192424282)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B C Yugesh

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. Cybersecurity

A. A cybersecurity system historically detects 95% of malicious activities. The new algorithm correctly detects 194 out of 200 attacks.

Step 1: Identify the population and sample

Population: All malicious activities that the cybersecurity system may receive.

Sample: 200 attacks tested using the new detection algorithm.

Correctly detected attacks = 194.

Step 2: Identify the parameter of interest

Let p be the true proportion of malicious activities correctly detected by the new algorithm.

Step 3: Calculate the point estimate

$$\hat{p} = x/n = 194/200 = 0.97.$$

Therefore, the estimated detection rate is 97%.

Step 4: State H_0

$$H_0: p = 0.95.$$

Step 5: State H_1

$$H_1: p > 0.95.$$

This is a right-tailed test because we are checking whether the detection rate has improved.

Step 6: Select the significance level

$$\alpha = 0.05.$$

Step 7: Calculate the test statistic

For a one-proportion z-test:

$$\begin{aligned} z &= (\hat{p} - p_0) / \sqrt{[p_0(1-p_0)/n]} \\ &= (0.97 - 0.95) / \sqrt{[(0.95)(0.05)/200]} \\ &= 0.02 / 0.01541 \\ &\approx 1.297. \end{aligned}$$

Step 8: Calculate the p-value

$$\text{p-value} = P(Z > 1.297) \approx 0.0973.$$

Step 9: Compare p-value with α

$$0.0973 > 0.05.$$

Therefore, we fail to reject H_0 .

2. Healthcare

A. Before implementation, the average patient waiting time was 42 minutes.

After implementation, 64 patients had a mean waiting time of 38 minutes and standard deviation 12 minutes.

Step 1: Identify the population and sample

Population: All patients using the new appointment scheduling system.

Sample: 64 patients observed after implementation.

Step 2: Identify the parameter of interest

Let μ be the true average patient waiting time after implementation.

Step 3: Calculate the point estimate

The point estimate of μ is the sample mean:

$$\bar{x} = 38 \text{ minutes.}$$

Step 4: State H_0

$$H_0: \mu = 42 \text{ minutes.}$$

Step 5: State H_1

$$H_1: \mu < 42 \text{ minutes.}$$

This is a left-tailed test because the objective is to determine whether waiting time has decreased.

Step 6: Select the significance level

$$\alpha = 0.05.$$

Step 7: Calculate the test statistic

$$\text{Standard error} = s/\sqrt{n} = 12/\sqrt{64} = 12/8 = 1.5.$$

$$\begin{aligned} z &= (\bar{x} - \mu_0)/(s/\sqrt{n}) \\ &= (38 - 42)/1.5 \\ &= -4/1.5 \\ &\approx -2.667. \end{aligned}$$

Step 8: Calculate the p-value

$$\text{p-value} = P(Z < -2.667) \approx 0.0038.$$

Step 9: Compare p-value with α

$$0.0038 < 0.05.$$

Therefore, we reject H_0 .

3. Social Media Analytics

A. A social media platform claims that the average daily usage time is 3 hours.

A random sample of 100 users has a mean usage time of 3.2 hours and standard deviation 0.8 hours.

Step 1: Identify the population and sample

Population: All users of the social media platform.

Sample: 100 randomly selected users.

Step 2: Identify the parameter of interest

Let μ be the true average daily usage time of all users.

Step 3: Calculate the point estimate

$\bar{x} = 3.2$ hours.

Step 4: State H_0

$H_0: \mu = 3$ hours.

Step 5: State H_1

$H_1: \mu \neq 3$ hours.

This is a two-tailed test because the claim is tested for any difference from 3 hours.

Step 6: Select the significance level

$\alpha = 0.05$.

Step 7: Calculate the test statistic

Standard error = $s/\sqrt{n} = 0.8/\sqrt{100} = 0.8/10 = 0.08$.

$z = (\bar{x} - \mu_0)/(s/\sqrt{n})$

$$= (3.2 - 3)/0.08$$

$$= 0.2/0.08$$

$$= 2.50.$$

Step 8: Calculate the p-value

For $z = 2.50$, one-tail probability ≈ 0.0062 .

Because this is a two-tailed test:

$$\text{p-value} = 2(0.0062) \approx 0.0124.$$

Step 9: Compare p-value with α

$$0.0124 < 0.05.$$

Therefore, we reject H_0 .

Step 10: Final conclusion

The data provide sufficient evidence at the 5% significance level to reject the company's claim that the average daily usage time is 3 hours. The observed average of 3.2 hours is significantly different from 3 hours.

4. Machine Learning

A. The existing classification model has an accuracy of 82%. The new model achieves 86% accuracy on a randomly selected test sample of 500 observations.

Step 1: Identify the population and sample

Population: All future observations on which the new classification model may be used.

Sample: 500 randomly selected test observations.

Step 2: Identify the parameter of interest

Let p be the true accuracy of the new classification model.

Step 3: Calculate the point estimate

$$\hat{p} = 0.86.$$

Thus, the observed accuracy of the new model is 86%.

Step 4: State H_0

$$H_0: p = 0.82.$$

Step 5: State H_1

$$H_1: p > 0.82.$$

This is a right-tailed test because we want to know whether the new model improves accuracy.

Step 6: Select the significance level

$$\alpha = 0.05.$$

Step 7: Calculate the test statistic

$$\text{Standard error under } H_0 = \sqrt{[p_0(1-p_0)/n]}$$

$$= \sqrt{[(0.82)(0.18)/500]}$$

$$\approx 0.01718.$$

$$z = (0.86 - 0.82)/0.01718$$

$$\approx 2.328.$$

Step 8: Calculate the p-value

$$\text{p-value} = P(Z > 2.328) \approx 0.0100.$$

Step 9: Compare p-value with α

$$0.0100 < 0.05.$$

Therefore, we reject H_0 .

5. Manufacturing / Quality Control

A. A manufacturer claims that the production process has a defect rate of only 2%. From 2,000 randomly selected products, 55 are defective.

Step 1: Identify the population and sample

Population: All products produced by the manufacturing process.

Sample: 2,000 randomly selected products.

Defective products in the sample = 55.

Step 2: Identify the parameter of interest

Let p be the true defect rate of the production process.

Step 3: Calculate the point estimate

$$\hat{p} = x/n = 55/2000 = 0.0275.$$

Therefore, the estimated defect rate is 2.75%.

Step 4: State H_0

$$H_0: p = 0.02.$$

Step 5: State H_1

$$H_1: p > 0.02.$$

This is a right-tailed test because we are checking whether the actual defect rate is greater than 2%.

Step 6: Select the significance level

$$\alpha = 0.05.$$

Step 7: Calculate the 95% confidence interval and test statistic

95% confidence interval for p:

$$\begin{aligned}\hat{p} \pm 1.96\sqrt{[\hat{p}(1-\hat{p})/n]} \\&= 0.0275 \pm 1.96\sqrt{[(0.0275)(0.9725)/2000]} \\&\approx 0.0275 \pm 0.00717 \\&\approx (0.02033, 0.03467).\end{aligned}$$

Therefore, the 95% confidence interval is approximately 2.03% to 3.47%.

For the hypothesis test:

$$\begin{aligned}z &= (\hat{p} - p_0) / \sqrt{[p_0(1-p_0)/n]} \\&= (0.0275 - 0.02) / \sqrt{[(0.02)(0.98)/2000]} \\&\approx 2.396.\end{aligned}$$

Step 8: Calculate the p-value

$$\text{p-value} = P(Z > 2.396) \approx 0.0083.$$

Step 9: Compare p-value with α

$0.0083 < 0.05$, so we reject H_0 .

Confidence interval check: The claimed value 0.02 (2%) is below the approximate lower limit of the 95% confidence interval, 0.02033 (2.03%).

This also supports rejection of the manufacturer's claim.